MKALAMA DISTRCT COUNCIL

FORM TWO MOCK ASSESSMENT

BIOLOGY

TIME: 2:30 Hours **MAY, 2025**

Instructions

1. This paper consists of Sections **A, B and C** with a total of **ten (10)** questions.
2. Answers all questions.
3. All writings must be in **blue** or **black** ink except for drawings which **must** be in pencil.
4. Calculator and cellular phone are **not** allowed in the examination room.
5. Write your **Assessment Number** at the top right corner of every page.

SECTION A (15 Marks)

Answer **all** questions in this section.

1. For each of items, **(i) – (x),** choose the most correct answer among the given alternatives and write its letter beside the item number in the box provided.

(i) Immunity is the ability of the body to defend itself against infections. Which one of the following types of immunity involves passage of antibodies from mother to unborn baby across the placenta?

A. Natural passive immunity
B. Artificial active immunity
C. Artificial passive immunity
D. Artificial active immunity

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(ii) A person picks up a telephone after hearing it ring. Which characteristics of living things is the person showing by this action?

A. Movement and locomotion
B. Movement and irritability
C. Growth and response
D. Development and response

(iii) Which apparati are used for heating substance in the biology laboratory?

A. Test tube holder and thermometer
B. Spirit lamp and test tube rack
C. Bunsen burner and Thermometer
D. Bunsen burner and Spirit burner.

(iv) When two organisms live together for their mutual benefit the relationship is called:

A. Commensalism
B. Mutualism
C. Epiphytism
D. Parasitism

(v) Conditions or factors that can change or be changed in an experiment

A. Hypothesis
B. Problems
C. Variables
D. Conclusions

(vi) The residents of a certain village in Iringa feed on meat, beans and stiff porridge. In order for their health to improve, which of the following would you advise them to add to their present diet?

A. Fish and sweet potatoes
B. Fruits and green vegetables
C. Water
D. Bread and green vegetables

(vii) All of the following are found suspended in the plasma except:

A. Red blood cells
B. Platelets
C. Capillaries
D. White blood cells

(viii) Why bryophytes are the most primitive plants on earth?

A. They grow on wet areas
B. They do not produce seeds
C. They reproduce asexually and sexually
D. They lack vascular tissues

(ix) People who are always talking while eating are most likely to be:

A. Highly respective
B. Express bad manner
C. Express good manners
D. Good role model to others

(x) Which of the following is among the products of an aerobic respiration in plants?

A. Ethanol
B. Lactic acid
C. Oxygen
D. Hydrogen

2. Match the phrase in **LIST A** with responses in **LIST B** by writing the letter of the correct response in a table provided.

LIST A	**LIST B**
(i) Heavy loss of blood	A. Thrombosis
(ii) Overreaction of body to antigen	B. Leukemia C. Agglutination
(iii) An excessive increase in white blood cells in the blood.	D. Allergy E. Haemophilia
(iv) Inability of blood to clot	F. Anaemia G. Haemorrhage
(v) Ability of blood cells to clump together	H. Hypertension I. Tonsillitis

SECTION B (70 Marks)

Answer **all** questions in this section.

3. (a) Mr. Amos asks you to prove that, first aid is important in the laboratory. (Provide **three** points).
(b) Differentiate the following terms:
(i) Pathogen and Vector

(ii) Obesity and Anorexia nervosa

(c) The health officers advise people to breathe through the nose instead of using the mouth. Give **three** reasons.

4. (a) A bread was put in a wet cupboard after few days black thread like structures which ended up in bulb like shape appeared on the bread.

(i) Write the common name of the organism that grew on the bread surface

(ii) Name the kingdom in which the organism belongs

(iii) Name the phyla in which the observed organism belongs

(iv) State **two** advantages of the kingdom into which the organism named in (a) above belongs**…….……and……**

(c) Tanzania Government has been insisting the provision of vaccine to its children under 5 years old. List **four** common diseases against which children are immunized.

(d) Write a word equation for aerobic respiration. **……**

5. A group of form II students from Gunda Secondary School had a study visit at Iambi Luteran Hospital in Mkalama. They noted a patient with sunken eyes, severe dehydration, weak pulse, deep breathing, vomiting and watery diarrhea like rice water.

(a) What could be the possible disease that the patient was suffering from? **……**

(b) Briefly explain **four** control or preventive measures of the disease mentioned in (a) above.

(c) Outline **four** differences between the digestive system of a cow and that of human.

6. (a) State the meaning of transpiration **……...**

(b) Giving **four** points, explain why transpiration is important to plants.

(c) The following terms are arranged in alphabetical order. Rearrange them into appropriate biological sequence, starting with the lowest level.

Cell → organ → organism → system → tissue

7. (a) Draw and give one function of each of the following biology laboratory apparatus

(i) Dropper

(ii) Hand lens

(c) Similar types of three animal cells were placed in 3 different solutions, named **A, B** and **C.** After some time, the cell in solution **A** bursted the cell in solution **B** shrank and the cell solution **C** did not change.

(i) Which type of solutions were **A, B, C**

(ii) Explain the reasons for such observations.

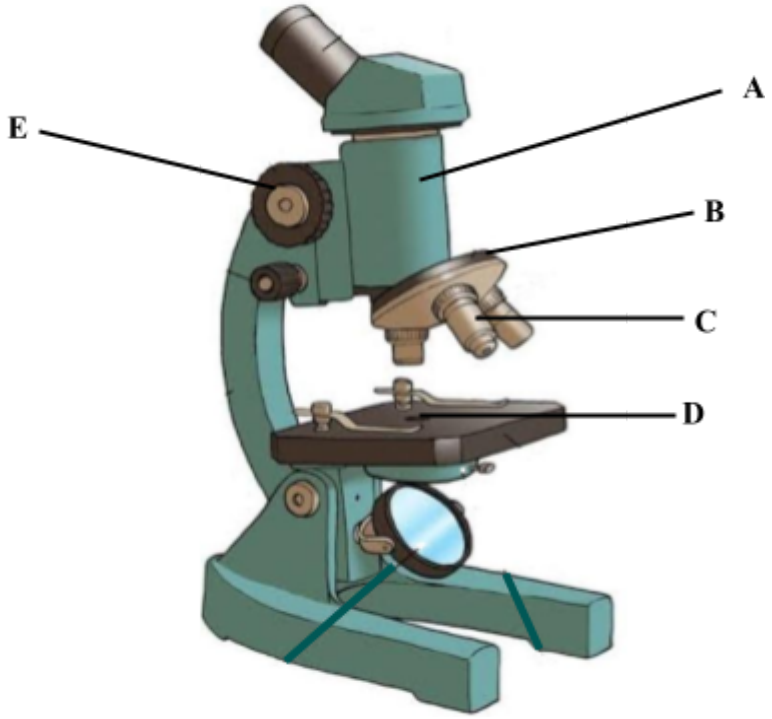
8. (a) On the way to school, Naomi saw the following materials along the road; broken glasses, ripe banana peelings, syringes, old newspapers and water from bathroom.

(i) Name the materials from the above list that can be recycled**……**

(ii) Give **one** effect of those materials in the place where they were placed**…….**

(iii) From the above list, name **one** example of clinical waste **……..**

(b) Name the parts of the microscope labelled **A - E** in the diagram below and state the function of each.



9. An ecologist carried out a survey to estimate the number of organisms in a certain dam. The following table shows the record of the survey:

Organisms	Estimated number
Microscopic algae	450,000
Small fish	200
Mosquito larvae	5,500
Crocodiles	10
Large fish	160

(a) Which of the above organisms are called the:

(i) Producer …**……**

(ii) Primary consumers **……**

(iii) Consumers of the last order **……**

(c) With one reason, name the organism which is likely to get finished first in the dam.
Organism**……**Reason **……..**

(d) Explain **four** disadvantages of classifying organisms using few observable features.

SECTION C (15 Marks)

10. POMBESICHAI got an accident when he was walking alone and when he was admitted to the hospital, he was advised by doctor to use a certain type of food for recovering and repairing his body.
    (a) Suggest the food type he was advised to use.
    (b) List down **three** natural sources of food type suggested in **10(a)**
    (c) Outline **three** roles of the food suggested above in **10(a)**
    (d) Identify and explain **three** effects and four symptoms of the disease which results when the individuals below five years old lack the food named in**10(a)** above in the diet.